

ABC INSTITUTE OF TECHNOLOGY
AUTONOMOUS EXAMINATIONS BRANCH, HYDERABAD
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
DAA TEST PAPER

Branch / Dept: Computer Science and Engineering	Course Code: R20-CSE	Subject Code: CS-501
Subject: Design and Analysis of Algorithms	Date of Exam: 15-11-2026	Max Marks: 15 Marks
Semester / Year: III Year II Semester	Academic Year: 2026-2027	Time / Duration: 3 Hours

General Instructions:

- 1. Answer all questions in Part A.
- 2. In Part B, answer either (a) or (b) from each question.
- 3. Assume suitable missing data if necessary.

PART A

CORE QUESTIONS

Instructions: Answer any 3 questions out of 5. [5 Questions × 5M | Attempt any 3 | Max Marks: 15]

1. Explain the fundamental working principle of Greedy Method with suitable illustrations. **[5 Marks]**
2. Illustrate the step-by-step mechanism of Matrix Chain Multiplication using a concrete example. **[5 Marks]**
3. Explain the fundamental working principle of Knapsack Problem with suitable illustrations. **[5 Marks]**
4. Compare and contrast the advantages and limitations of Longest Common Subsequence.. **[5 Marks]**
5. How is Job Sequencing. implemented in modern software systems? Explain with pseudocode or schema. **[5 Marks]**